Status	Finished
Started	Sunday, 18 October 2026, 12:13 PM
Completed	Sunday, 18 October 2026, 12:19 PM
Duration	5 mins 14 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Heap Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It sorts data using a single pass
- b. It is less efficient than bubble sort
- c. It does not require recursion
- d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Merging
- c. Complexity
- d. Sorting

The correct answer is:
Complexity

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It divides the list into sublists
- c. It uses the heap data structure
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **9**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort

b. Quick Sort

c. Merge Sort

d. Insertion Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort

b. Quick sort

c. Selection sort

d. Merge sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Dividing the list into halves
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Linear Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Provides a basis for other algorithms to work efficiently
c. Makes data harder to manage
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently